

CLARK ABOUTRJEILY

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EDUCATION

Georgia Institute of Technology <i>Atlanta, GA</i>	08/2024 – 05/2026
- M.S. in Robotics, with Core Areas in Controls, Artificial Intelligence (AI), and Perception, GPA 4.0/4.0 - Coursework: Robotics, Nonlinear & Networked Control, Deep Reinforcement Learning, Computer Vision	
American University of Beirut (AUB) <i>Beirut, Lebanon</i>	08/2020 – 06/2024
- B.Eng. in Electrical and Computer Engineering with a Minor in Business Administration, GPA 4.0/4.0 - Graduated with High Distinction, with a Track in Control, Robotics, and Intelligence Systems, ranked 1st - Coursework: Autonomous Mobile Robotics, Machine Learning, Modern Control Systems, Instrumentation	

EXPERIENCE

FACTS Lab, Georgia Tech, Research Assistant <i>Atlanta, GA</i>	05/2025 – Present
- Integrating 6-DOF industrial manipulators for ROS2-based multi-agent collaboration and VLM-driven tasks - Enabling safety-critical bimanual planning and control with Signal Temporal Logic (STL) constraints	
STAR Lab, Georgia Tech, Research Assistant <i>Atlanta, GA</i>	01/2025 – 04/2025
- Developed a motion planner using nonlinear optimization for heterogeneous multi-robot coordination - Added a coalition-aware abstraction to improve synchronization control and eliminate inter-robot collisions	
American University of Beirut (AUB), Teaching Assistant <i>Beirut, Lebanon</i>	09/2022 – 05/2024
- Cooperated with professors in teaching Electronics I & II to junior ECE students - Enhanced concept delivery and graded weekly assignments and quizzes for 50+ students	
Vision and Robotics Lab (VRL), AUB, Research Intern <i>Beirut, Lebanon</i>	06/2023 – 08/2023
- Generated a novel multi-sensor, asynchronous, real, and collaborative 3D reconstruction dataset - Conducted motion capture, experiment design, and sensor fusion using C#, OptiTrack Motive, Cyclone 3DR - Co-authored a research paper for IROS, 2nd author, Didymos-XR Project	
L'Art de la Musique (School of Music), Teacher <i>Beirut, Lebanon</i>	09/2022 – 08/2023
Gave Piano, Music Theory, and Aural Skills courses to students of ages 6 to 30+	

PROJECTS

Look-ahead Unified Camera-LiDAR Inference and Diffusion (LUCID) <i>Atlanta, GA</i>	09/2025 – Present
Built a multimodal DiT pipeline for dynamics-conditioned future 3D LiDAR generation on nuScenes data	
Implementation and Adaptation of Diversity Is All You Need (DIAYN) <i>Atlanta, GA</i>	02/2025 – 04/2025
- Reproduced and validated the base DIAYN algorithm across multiple benchmarks (e.g. hopper, cheetah) - Extended DIAYN to a 6-DOF Panda Arm, augmented with HRL/VAE to learn more human-relevant skills	
Decentralized Heterogeneous Ant Colony Optimization (ACO) <i>Atlanta, GA</i>	02/2025 – 04/2025
- Built a decentralized ACO framework for multi-agent cooperation on Georgia Tech's Robotarium - Integrated reinforcement learning to adaptively balance exploration and connectivity for faster convergence	
The Robot Collective: Traffic-Aware Multi-Agent Path Planning, Georgia Tech VIP <i>Atlanta, GA</i>	08/2024 – 12/2024
- Improved simulator efficiency of graph-based planner with dynamic edge weighing for block-stacking bots - Mentored undergraduate team members to familiarize them with ROS and to enhance workflow	
Mixed Reality Virtual Testbed for HoloLens-Robot Multi-Agent System <i>Beirut, Lebanon</i>	09/2023 – 05/2024
- Developed a Unity3D AR application for human-robot collaborative operation through ROS 2 and MoveIt - Augmented the physical mobile robot with a virtual robotic manipulator to cut simulation costs	
SLAM, Kinematics and Motion Planning for Ground and Aerial Vehicles using ROS2 <i>Beirut, Lebanon</i>	10/2023 – 12/2023
- Devised an occupancy grid mapping and path planning algorithm for differential drive robot navigation - Developed the autonomous motion control model of a 6-DOF drone	

EXTRACURRICULAR ACTIVITIES

L'Art de la Musique (School of Music) <i>Beirut, Lebanon (Currently Remote, Atlanta, GA)</i>	01/2007 – Present
- Level VIII of Piano Performance, certified by The Lebanese National Higher Conservatory of Music - Advanced Music Theory, Aural Skills, Music History, Ethnomusicology	
AUB Robotics Club (AUBRC), Member <i>Beirut, Lebanon</i>	10/2020 – 05/2024
Assisted and participated in several workshops for ROS, Arduino, and Raspberry Pi	
Build It Weekend 5.0 AUBRC x IEEE AUB Student Branch, Robotics Competition <i>Beirut, Lebanon</i>	03/2023 – 04/2023
- Won 2nd place in a 48-hour robotics competition out of 10 teams, judged by AUB professors - Designed and built a prototype lifeline rescue-bot with IoT control using ESP32	
Diverse Volunteering <i>Beirut, Lebanon</i>	12/2016 – 01/2020
Provided food, aid and entertainment for elderly homes, childcare centers, and BETA Lebanon's dog shelter	

SKILLS AND INTERESTS

Languages	Fluent in English, French and Arabic – Basic Korean (writing, reading, speaking)
Hard Skills	MS Office, LaTeX, GitHub, Python, PyTorch, C#, C++, ROS, ROS2, Gazebo, RViz, MATLAB, LabVIEW, PSpice
Soft Skills	Time Management, Communication, Problem-solving, Critical Thinking, Teamwork, Collaboration
Interests	Tennis, Ski, Piano, Robotics, Travel